

Mitsue Project — Business Case & Decision Framework

The "Why Say Yes" Layer

This document sits **above** the [Implementation Plan](#). The implementation plan answers "what we will build and when." This document answers the question every decision-maker actually asks: **"Why should I say yes, and what do I get?"** It consolidates the financial and decision logic scattered across the revenue model, funding flowchart, EVM, and risk register, and fills the three gaps those documents leave open: (1) the entity & financing structure, (2) the alternatives the village is really choosing between, and (3) whole-project return, residual risk, and decision authority.

All financial figures are illustrative early-stage estimates consistent with [mitsue revenue model.md](#) and [mitsue evm plan.md](#). Phase 1 feasibility studies (due M9, Dec 2026) replace them with vetted figures.

1. Entity & Financing Structure — what each funder gets

The project raises money three ways — **grants, equity, and revenue/energy-share (no debt)**. These cannot all live in one organisation: a Japanese non-profit (一般社団法人 / NPO法人) **cannot issue equity or distribute profit to investors**. The project therefore uses a deliberate **two-tier structure**, which is also its mission-protection mechanism.

1.1 The two tiers

	Tier 1 — Mission entity	Tier 2 — Operating company
Form	一般社団法人 (non-profit)	合同会社 GK (for-profit)
Holds	Charter, 官民連携 relationship with village, brand, IP/"playbook", forestry-restoration programme, J-Credit registration	Data-center assets, energy plant (biomass CHP, solar, battery), grid/FIT contracts, hosting contracts, staff
Receives	Founder capital, government grants, foundation grants — all non-distributing	Equity investment, revenue/energy-share contracts, operating revenue
Controls	Owns the majority of the GK + holds a charter "mission-lock" clause	Operated under contract to the mission entity
Returns to funders	None — return is the mission (public benefit)	Dividends / revenue-share from operating surplus

The mission entity owns and controls the operating company. Grant money (which legally cannot generate private profit) stays in Tier 1 or funds shared infrastructure; investor money that expects a return sits in Tier 2, where returns are legal and contractual. The 交付金 (MoE decarbonization grant) flows to the **village**, which co-invests the subsidized capex into the shared energy assets via the 官民連携 partnership.

1.2 What each funding layer is, and what it gets back

Layer	Source	¥M	Instrument	What they get
L1	Founders (Rob + JP co-founder)	6	Founder equity in GK + sweat	Minority equity; mission stewardship
L2	Government grants (national/pref./village) incl. 交付金 2/3-3/4 capex subsidy	115	Non-refundable grant (to village / Tier 1)	RE-plan targets met; decarbonization outcomes; no financial return expected
L3	Foundations	33	Mission-restricted grant	SDG/impact outcomes; reporting; brand association
L4	Corporate partners	35	Choice of: CSR grant · minority GK equity · prepaid hosting (revenue advance) · energy-share offtake	Green-compute capacity, ESG story, or contracted return depending on instrument chosen
L5	Operating revenue	3+	Reinvested surplus	—
Gap	To close ¥28-53M	28-53	Additional 交付金 + corporate GK equity + revenue-financing	As per L2/L4 above

1.3 Mission-lock — why investors cannot capture the project

Because equity exists only in Tier 2 and the Tier-1 non-profit holds the majority plus a charter clause, investors get a **real but minority** return and **cannot redirect the mission, sell the forest programme, or strip the community assets**. This is the answer to "what am I committing to / what do you promise": grant funders commit

to a public good and are promised outcomes; equity/revenue-share investors commit capital to Tier 2 and are promised a contracted share of a defined operating surplus — bounded so the village always retains control.

1.4 Return profile (illustrative, base case)

Operating surplus accrues in the GK: **net surplus ¥10–32M/yr by Year 5**, project-level capital payback **10–18 years**. Equity-investor return depends on entry valuation and share, both set in Phase 2 once feasibility figures exist — **this document defines the *structure*; Phase 2 sets the *price***. No instrument promises a return the operating surplus cannot support.

2. Alternatives Analysis — what the village is really choosing between

Every decision-maker has a "do nothing" option. For the village it is concrete: **underused space stays underused**. The project will choose its premises from **several candidate sites**, all underused village building stock. One strong candidate is the **Koryukan (みつえ体験交流館 / Mitsue Experience Exchange Center)** — a former wooden elementary school that today operates as a cultural and educational facility but has empty, underused classrooms; here the hub would occupy spare capacity and **complement** the Koryukan's existing crafts-and-experience role rather than replace it. Disused former factory or workshop buildings are other candidates. Final site selection is a Phase 1 decision.

Note on the dome school: the dome school is a *separate* building, already in its own bidding/selection process to be reused by another party — it is **not** part of this project and should not be confused with the Koryukan.

A proposal is only persuasive against the alternatives it beats. For the candidate premises the village is realistically choosing among five paths:

Option	Upfront cost to village	Ongoing	Jobs	Energy resilience	Forest / ecology	Productive reuse	Unlocks 交付金	Reversible
A. Status quo — spare capacity idle	¥0 now	Ongoing upkeep, no new return	0	None	Forest untended (liability grows)	No new use	No	Yes
B. Solar-only	Low-med capex	FIT income, thin	~0	Intermittent only	None	Partial	Partly (no resilient site)	Yes
C. Forestry-only	Low	J-Credit + timber, slow	Few	None	Restored ✓	No	No	Yes
D. Sell / lease to another party	One-time cash or rent	Control lost	Uncertain	None	Lost	Buyer's choice ¹	No	Largely no
E. Integrated project (proposed)	Mostly grant-funded ²	Net surplus by ~Y5	8–15	24/7 (biomass CHP island-mode)	Restored ✓	Yes ✓	Yes ✓	Staged / descopeable

¹ Where a private buyer is involved, the village should weigh buyer credibility — the dome-school bidding file shows prospective-buyer quality varies; selling or leasing does not guarantee a good village outcome. ² Village cash exposure is minimized because the 交付金 subsidizes 2/3–3/4 of eligible energy capex and other layers carry the rest.

Why the integrated project wins

- **It is the only option that satisfies the village's own published RE plan.** That plan's "one resilient renewable + storage + EV site" indicator currently reads zero; only Option E provides the "1". Options A–D leave the village's adopted 2050 strategy without an operator.
- **It is the only option with an anchor offtaker.** Rural energy projects in Japan typically fail their financials for lack of steady demand. The data center is the 24/7 baseload that makes the energy economics — and therefore the EV charging and blackout resilience — actually close. Solar-only and forestry-only have no anchor and stall.
- **It puts underused space to productive use.** Option A keeps paying upkeep on empty classrooms; Option E turns spare capacity into a revenue-generating facility (and, at the Koryukan, complements its cultural role).
- **It is the only option that does several jobs at once** — heals the forest, generates energy, creates higher-skill local jobs, and produces a replicable model — because the thinnings that fund the energy are the same thinnings that restore the forest.

Honest counter-point: Option E is also the **highest in cost, complexity, and execution risk**. The mitigant is structural, not rhetorical — the **funding-gate / descope mechanism** (§4) means the village never commits to the full build up front; each phase proceeds only if the prior gate's funding and feasibility clear. If the project stalls, it stops at a gate having delivered partial value, not at an idle building.

3. The ask, the offer, the proof — per audience

Each decision-maker needs the same underlying story ordered for *their* question: **Why you?** → **What I'm asked for** → **What I get** → **How I know you'll deliver**.

Audience	What we ask	What they get	How they can be confident
Village government	Lease premises (Koryukan classrooms / factory — one of several candidate sites); endorse as 官民連携 partner; co-apply for 交付金	Depopulation reversal, 8–15 jobs, RE-plan delivered, underused space activated	Staged gates (no lock-in); open data; feasibility studies before any build
MoE / grant bodies	Award 交付金 / planning grants	Adopted RE plan actually implemented; carbon; a replicable 過疎 model	Plan already adopted (step 1 done); milestone-based disbursement
Equity investors (Tier 2)	Minority capital into the GK opco	Contracted share of operating surplus (net ¥10–32M/yr by Y5); green-compute exposure	Two-tier structure; anchor off-taker; payback model; descope safety
Corporate partners	Hosting offtake / CSR / equity	Cheap green compute, ESG story, or contracted return	Working pilot before scale; named advisory bench
Foundations	Mission-restricted grant	Measurable SDG/impact outcomes, strong narrative	Open methodology; transparent reporting
Forestry co-op / landowners	Sugi harvesting access; restoration participation	Fair harvest compensation; J-Credit income; reduced wildlife damage	Phase-1 contracts; village backing

Proof of delivery capability (the "how do I know" answer, assembled): founder's Safecast track record; named advisory bench (Joi Ito, Ray Ozzie); the village's *already-adopted* RE plan; the open-data / open-methodology commitment; and the gate structure that makes every commitment conditional and reversible.

4. Whole-project return, sensitivity & go/no-go

4.1 Scenario P&L by Year 5 (illustrative)

	Low	Base	High
Annual revenue	¥28M	¥45M	¥67M
Operating cost (~60%)	¥18M	¥27M	¥35M
Net surplus	¥10M	¥18M	¥32M
Payback (¥200–290M capital)	~18 yr	~14 yr	~10 yr

4.2 The three swing variables (what would break the ROI)

The return is most sensitive to three figures, all resolved in Phase 1–2 feasibility:

- Data-center occupancy / hosting price** — the anchor. If hosting demand is weak, the whole stack thins.
Mitigant: secure a corporate offtake LOI before Gate 3 build.
- Biomass CHP feedstock cost & sizing** — if thinning logistics cost more than modelled, energy margin erodes.
Mitigant: independent feasibility early; forestry programme supplies fuel at internal cost.
- FIT/FIP tariff outcome** — sets surplus-energy value. *Mitigant: revenue is not dependent on FIT alone; it is one of six streams.*

4.3 Go/no-go is already built in

The project's decision spine **is** a go/no-go system — it is just not labelled as one. Each gate is a whole-project kill/continue point:

Gate	Funding test	If short
G1 (after Phase 0)	¥3–8M secured	Hold & re-pitch
G2 (after Phase 1)	¥30–50M + feasibility clears	Hold / descope
G3 (after Phase 2)	¥120–290M + offtake LOI	Stage the build
G4 (after Phase 3)	Revenue online?	Partial → stay in pilot

Kill criteria beyond funding: no JP co-founder verbal commitment by end of Phase 0 → G1 held; building structurally unfit (Phase 1 survey) → redesign or stop; feasibility shows negative base-case surplus → descope to forestry+solar only.

5. Residual risk & key dependencies

The [implementation plan risk register](#) lists 9 risks with mitigations. What it omits — and investors ask for — is the **residual** (what remains after mitigation) and a consolidated **dependency map**.

5.1 Residual risk (after mitigation)

Risk	After mitigation, what remains
Founder dependency (Rob)	Reduced by funded Rep.-Director role + Article 8a succession, but a single-person loss in Phase 0–1 still sets the project back months. Unmitigated tail.
JP co-founder not secured	Gate-1 hold protects spend, but failure to find one <i>ends</i> the project. Hard dependency.
Hosting demand weak	Six revenue streams cushion it, but the anchor thesis weakens; project survives smaller, ROI lengthens.
Feedstock economics	Internal-cost fuel helps, but persistent high logistics cost permanently compresses energy margin.

5.2 Key dependencies (in order)

1. **JP co-founder** — gates everything (legal entity, local trust, bus-factor).
2. **Village endorsement** — unlocks 交付金 and the building.
3. **交付金 award** — closes most of the funding gap.
4. **Corporate offtake LOI** — validates the anchor before the expensive build.
5. **Fiber/grid connection** — physical feasibility.

5.3 Key personnel & bus-factor

Currently the team's load-bearing point is the founder. Mitigations in place: funded Representative-Director role (removes unpaid-time dependency), Article 8a continuity/succession (interim Rep. Director appointed within 30 days of a key founder's loss; credentials and documented processes maintained). **Remaining exposure:** the JP co-founder seat is unfilled, and no second operational person yet exists — closing both is the top Phase-0 priority. (A *short positive team-profile* — who the principals are and why they are credible to investors — should be added once the JP co-founder is named.)

6. Decision table — authority + investor evaluation

Two linked views: **who decides what (authority)**, and **what each decision-maker needs to see (evaluation)**.

Decision	Authority	Gate / when	Evidence they need	What they get if yes
Lease/endorse premises (Koryukan / factory — site TBD Phase 1)	Village council / mayor	Phase 0–1	Alternatives table; job forecast; site survey	Revitalization; RE-plan delivery; idle asset revived
Award 交付金	MoE (via village)	Phase 1–2	RE-plan fit; carbon; 過疎 eligibility	Adopted plan implemented
Commit as JP co-founder	The individual	Phase 0 (G1 trigger)	Charter; role; compensation (funded)	Founding equity + mission role
Invest equity in GK	Investor	Phase 2 (G3)	Two-tier structure; P&L; sensitivity; offtake LOI	Contracted minority return
Commit hosting offtake	Corporate partner	Phase 2–3	Pilot performance; price/kWh; green cert	Cheap green compute / ESG
Supply feedstock	Forestry co-op / landowners	Phase 1	Harvest terms; J-Credit share	Forest income; reduced crop damage
Proceed past each gate	Board (Tier 1)	G1–G4	Funding secured + feasibility/kill-criteria	Authorisation to spend next phase

Sources & status

Financial ranges consolidate the project's own illustrative estimates from [mitsue revenue model.md](#), [mitsue implementation plan.md](#) §ROI, and [mitsue evm plan.md](#) — not externally audited. 交付金 terms: MoE 地域脱炭素移行・再エネ推進交付金, 実施要領 (補助率 2/3・3/4) <https://www.env.go.jp/content/900470616.pdf>. Entity structure (一般社団法人 + 合同会社 GK) per [mitsue implementation plan.md](#) §1A and the NGO-setup checklists.

Items still to confirm: final site selection from the candidate premises (Koryukan spare classrooms and/or a disused factory — Phase 1; the dome school is a separate building, out of scope); equity entry valuation (Phase 2); positive team profile (post JP co-founder).